



FINAL EXAM REVIEW SHEET – MATH 102A

- The final exam will be held online on **Saturday, June 10, 12:00 AM – 01:30 PM.**

Question One: Find where the following functions are continuous:

1. $f(x) = \sqrt{4-x^2} + \ln(1-x)$ 2. $f(x) = \frac{x}{\log_3(x+1)-2}$ 3. $f(x) = \frac{x}{1+\cos(x)}.$

Ans. 1. $[-2, 1)$ **2.** $(-1, 8) \cup (8, \infty)$ **3.** $\mathbb{R} - \{\dots, -3\pi, -\pi, \pi, 3\pi, \dots\}.$

Question Two: Find the following limits:

1. $\lim_{x \rightarrow -2} \frac{x^2 + 3x + 2}{x^2 - x - 6}$ **Ans.** $\frac{1}{5}$

2. $\lim_{x \rightarrow 3} \frac{2 - \sqrt{x+1}}{x^2 - 9}$ **Ans.** $-\frac{1}{24}$

3. $\lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{4} + h\right) - \frac{1}{\sqrt{2}}}{h}$ **Ans.** $\frac{1}{\sqrt{2}}.$

4. $\lim_{h \rightarrow 0} \frac{(1+2h)^5 - 1}{h}$ **Ans.** 10.

Question Three: Find the equation of the tangent line to the graph of $f(x) = x \cos(x)$ at $x = \pi$.

Ans. $y = -x.$

Question Four:

• If $\int_0^2 2f(x) dx = 4$ and $\int_3^0 f(x) dx = 3$, find $\int_2^3 f(x) dx.$ **Ans.** -5

• If $\int_0^1 f(2x+1) dx = 3$, find $\int_1^3 f(x) dx.$ **Ans.** 6

• If $\int_0^1 f(x) dx = -1$ and $f(1) = 3$, find $\int_0^1 x f'(x) dx.$ **Ans.** 4

• If $\int_0^a (2x+1) dx = 2$, find all values of $a.$ **Ans.** 1 and -2

Question Five: Find the derivative of the following function:

1. $f(x) = \frac{x^2 - x}{\sqrt{x}}, \quad x = 1$ **Ans.** 1

2. $f(x) = e^{-x} \tan(x), \quad x = \frac{\pi}{4}$ **Ans.** $e^{-\frac{\pi}{4}}$

3. $f(x) = \sin^2(x^3)$ **Ans.** $f'(x) = 6x^2 \sin(x^3) \cos(x^3)$

4. $f(x) = \sqrt[3]{e^x \sin(x)} + 8, \quad x = 0$ **Ans.** $\frac{1}{12}$

5. $f(x) = \ln\left(\frac{e^{\cos(2x)}}{\sqrt{x^2+2}}\right),$ **Ans.** $f'(x) = -2 \sin(2x) - \frac{x}{x^2+2}$

6. $F(x) = \int_0^{x^2} \frac{t}{t^2+1} dt, \quad x = 1$ **Ans.** 1

7. $F(x) = \int_{\sin^2(x)}^0 \frac{1}{t+1} dt, \quad x = \frac{\pi}{4}$ **Ans.** $-\frac{2}{3}$

Question Six: Find the following integrals:

1. $\int_0^1 \frac{x^2 - x + 1}{\sqrt{x}} dx$ **Ans.** $\frac{26}{15}$
 2. $\int_0^2 \sqrt{4 - x^2} dx$ **Ans.** π
 3. $\int \frac{x}{\sqrt[3]{x^2 + 1}} dx$ **Ans.** $\frac{3}{4}(x^2 + 1)^{\frac{2}{3}} + C$
 4. $\int \sec^2(x) e^{\tan(x)} dx$ **Ans.** $e^{\tan(x)}$
 5. $\int \frac{2}{x \ln(x)} dx$ **Ans.** $2 \ln |\ln(x)| + C$
 6. $\int y \sin(y) dy$ **Ans.** $-x \cos(x) + \sin(x) + C$
 7. $\int e^{\sqrt{x}} dx$ **Ans.** $2e^{\sqrt{x}}(\sqrt{x} - 1) + C$
 8. $\int \frac{\ln(x)}{\sqrt[3]{x^2}} dx$ **Ans.** $3\sqrt[3]{x}(\ln(x) - 3) + C$
 9. $\int \frac{3x}{x^2 + x - 2} dx$ **Ans.** $\ln |x - 1| + 2 \ln |x + 2| + C$
 10. $\int_1^e \frac{1}{x(2 + \ln(x))(3 + \ln(x))} dx$ **Ans.** $\ln \frac{9}{8}$
-

Question Seven: Find the area of the region enclosed by the curves:

1. $y = x^2 + 1$ and $y = x + 3$ **Ans.** $\frac{9}{2}$
2. $y = |x|$ and $y = 2$ **Ans.** 4
3. $y = \sin(x)$ and $y = \cos(x)$ on $[0, \pi]$ **Ans.** $2\sqrt{2}$
4. $y = x^3$ and $y = \sqrt{x}$ **Ans.** $\frac{5}{12}$